

APPOINTMENTS

Postdoctoral Researcher , The University of Tokyo <i>Department of Computer Science, IGARASHI Laboratory & TACHI Laboratory</i>	Sep '25 to Current Tokyo, JP
Postdoctoral Researcher , Swiss Federal Institute of Technology Lausanne (EPFL) <i>Department of Computer Science, Geometric Computing Laboratory (GCM)</i>	Feb '25 to May '25 Lausanne, CH

EDUCATION

Swiss Federal Institute of Technology Lausanne (EPFL) <i>Ph.D in Computer Science – Major in Artificial Intelligence (GPA: 5.4/6.0)</i> • Relevant courses: Learning Theory, Optimization for Machine Learning, Information Theory, Distributed Information Systems	Sep '19 to Jan '25 Lausanne, CH
University of California Berkeley <i>Master of Science in Mechanical Engineering – Major in Fluids (GPA: 4.0/4.0)</i> • Relevant courses: Machine Learning, Optimization Methods, Tensor Calculus, Differential Geometry, Advanced Fluid Mechanics, Ocean Engineering	Aug '17 to Dec '18 Berkeley, CA
Institut Supérieur de l'Aéronautique et de l'Espace, Supaéro (ISAE Supaéro) <i>Equiv. Bachelor of Science and Master of Science in Aeronautical and Aerospace Engineering ("Ingénieur ISAE Supaéro", GPA: 4.0/4.0)</i> • Relevant courses: Advanced Probability and Statistics, Computer Science, Continuum Mechanics, Fluid Mechanics	Aug '15 to Sep '19 Toulouse, FR

RESEARCH EXPERIENCE

Google Research (Arkadia) <i>Research Intern (qbecker@) – Host: Dr. Urs Bergmann (ursbergmann@)</i> • Developed a differentiable geometric primitives composition module (JAX) • Programmed deep vision models (JAX, TF1, and TF2) that learn to decompose an occupancy mask into primitives • Implemented optimal transport-based losses to train models to simplify building footprints into primitives (JAX)	Jun '22 to Sep '22 Berlin, DE
EPFL <i>Ph.D Candidate – Advisor: Prof. Dr. Mark Pauly</i> • Introduced and implemented a rationalization algorithm for bending-active structures that optimizes a single kit of parts to approximate many user-defined designs • Implemented an inverse design pipeline for deployable assemblies of curved elastic beams (C++ with Python bindings) • Developed a forward design tool based on conformal map for C-shells (Rhino-Grasshopper plugin) • Designed generative models to solve constrained physics-based inverse problems • Developed differentiable physics simulation frameworks: billiard game with diverse obstacles (PyTorch), constrained elastic deformations of 3D volumetric objects (C++, PyTorch, and JAX)	Sep '19 to Jan '25 Lausanne, CH
Dassault Systèmes <i>Research Intern in Machine Learning</i> • Developed a clustering algorithm based on hash tables to find geometrically similar 3D parts within a dataset	Jan '19 to Jul '19 Paris, FR
UC Berkeley <i>Graduate Student – Advisor: Prof. Dr. Reza Alam</i> • Developed an ad hoc genetic algorithm coupled with a boundary element method to optimize underwater vehicles • Trained a deep neural network to morph an underwater vehicle's shape according to its environment	Sep '17 to Dec '18 Berkeley, CA
ISAE Supaéro <i>Undergraduate Student – Advisor: Prof. Dr. Laurent Joly</i> • Coded the method of characteristics for simulating supersonic flows and design nozzles, validated with CFD	Jan '17 to Jul '17 Toulouse, FR

PUBLICATIONS

Quentin Becker, Maria Larsson, Takeo Igarashi. Helical Box Joints. *SIGGRAPH Posters '26: Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters*.

Klara Mundilova, Michele Vidulis, **Quentin Becker**, Florin Isvoranu, Mark Pauly. C-Nets: An Interactive Design Framework for Cone-Nets. *Advances in Architectural Geometry 2025*.

Quentin Becker*, Oliver Gross*, Mark Pauly (*joint first authors). Inverse Geometric Locomotion. *ACM Transactions on Graphics (Proc. of SIGGRAPH 2025)*, 44.4, article 141 (August 2025): 1-17.

Michele Vidulis*, Klara Mundilova*, **Quentin Becker***, Florin Isvoranu, Mark Pauly (*joint first authors). C-Tubes: Design and Optimization of Tubular Structures Composed of Developable Strips. *ACM Transactions on Graphics (Proc. of SIGGRAPH 2025)*, 44.4, article 154 (August 2025): 1-19. (**Best Paper Award Honorable Mention**)

Quentin Becker*, Uday Kusupati*, Seiichi Suzuki, Mark Pauly (*joint first authors). Computational Design of a Kit of Parts for Bending-Active Structures. *ACM Transactions on Graphics (Proc. of SIGGRAPH Asia 2024)*, 43.6, article 230 (December 2024): 1-16.

Quentin Becker, Seiichi Suzuki, Mark Pauly. Interactive Design of C-shells Using Reduced Parametric Families. *Proceeding of LASS Annual Symposia* (20), 1-10.

Quentin Becker, Seiichi Suzuki, Yingying Ren, Davide Pellis, Julian Panetta, Mark Pauly. C-shells: Deployable Gridshells with Curved Beams. *ACM Transactions on Graphics (Proc. of SIGGRAPH Asia 2024)*, 42.6, article 181 (December 2023): 1-17. (**Best Paper Award Honorable Mention**)

Michelis, Mike Yan, and **Quentin Becker**. On Linear Interpolation in the Latent Space of Deep Generative Models. *ICLR 2021 Workshop on Geometrical and Topological Representation Learning*. 2021. (**Spotlight**)

Quentin Becker, Mohammad-Reza Alam, and Alexandre Immas. Hydrodynamic Design of a Morphic Autonomous Underwater Vehicle Using Neural Networks. *ASME 2019 38th International Conference on Ocean, Offshore and Arctic Engineering*. American Society of Mechanical Engineers Digital Collection. 2019.

INVITED TALKS

SNU, “Symmetry-Determined Motion: From Locomotion to Joinery”, hosted by Prof. Young Min Kim August 2026

KAIST, “Symmetry-Determined Motion: From Locomotion to Joinery”, hosted by Prof. Minhyuk Sung August 2026

AsiaGraphics Webinar, “Geometry-Informed Inverse Design for Architecture and Robotics”, hosted by Prof. Minhyuk Sung April 2026

HKUST, “Geometry-Informed Inverse Design for Architecture and Robotics”, hosted by Prof. Ziqi Wang December 2025

NVIDIA Toronto AI Lab, “Geometry-Informed Inverse Design of Physical Systems”, hosted by Prof. David Levin February 2025

TEACHING EXPERIENCE

Geometry for Digital Fabrication and Architecture, Summer School at DTU Copenhagen, DK
Instructor of “Elastic Rod Assemblies: From Geometric Principles to Physics-Based Design Optimization” Summer 2026

- Composed two lectures (three hours each) introducing concepts from differential geometry, elasticity, optimization. and inverse design.
- Created a tutorial codebase with exercises. The resulting design tool allows tracing asymptotic lines, and generating fabrication data for laser-cut asymptotic gridshells.
- Laser cut some prototypes designed by participants, with 3D scanning and comparison against the input surface.

Summer Geometry Initiative, Summer Research Program at MIT Boston, MA
Instructor (one week project) of “Motion Planning for Broken Snake Robots” Summer 2025

- Composed lecture slides for an introduction to geometric mechanics and control problems in this context.
- Created a tutorial codebase along with coding homework (geometric mechanics, adjoint sensitivity analysis).

EPFL Lausanne, CH
Teaching Assistant for CS-457 Geometric Computing Fall 2021, 2023

- Developed recitations, created theory and coding homework (FEM, autodiff, adjoint sensitivity analysis).

Co-Head Teaching Assistant for CS-341 Introduction to Computer Graphics Spring 2019, 2020

- Created coding homeworks (raytracing on the GPU), developed and led recitations, supervised coding projects.

Teaching Assistant for Math-101 Analysis I and II Fall 2020, Spring 2021

Teaching Assistant for CS-107 Introduction to Programming Fall 2022, 2024

UC Berkeley Berkeley, CA
Graduate Student Instructor (50%) for Physics-8A Introductory Physics Fall, Spring 2017

MENTORSHIP

PhD Projects

- Clément Jambon (PhD student, MIT); Topic: Soft-Body Lagrangian Mechanics Summer 2026
- Yinfei Qian (PhD student, UTokyo); Topic: Fabric Scrap Recomposition Spring 2026
- Mingi Lee (PhD student, SNU); Topic: Strict Contact Assembly Generation Spring 2026

Master Thesis

- Wataru Nakayama (MS student, UTokyo); Topic: Structural Evaluation of Wooden Integral Joints Spring 2026
- Orfeas Liossatos (MS student, EPFL); Topic: Inverse Geometric Propulsion Spring 2025

Semester Projects

- Mathilde Simoni (MS student, EPFL); Topic: Neural Subspaces for Symplectic Physical Trajectories Spring 2024
- Danila Zubko (MS student, EPFL); Topic: Latent Space Physical Simulations Fall 2023
- Vishal Pani (MS student, EPFL); Topic: Generative Model Evaluation Metric Using Differential Geometry Spring 2022

- Cosme Jordan (MS student, EPFL); Topic: Generative Inverse Design of Kirigami Sheets Fall 2021
- Amine Chaouchi (MS student, EPFL); Topic: Unsupervised Disentanglement of Caricatures Generation Spring 2020
- Mike Jan Michelis (MS student, TUM); Topic: Interpolations in a Generative Model's Latent Space Fall 2020
- Nathan Greslin (MS student, EPFL); Topic: Body Capture from a Single Image Fall 2020

Summer Interns

- Yuyan Wu (BS student, Brown); Topic: Design Interface for Traditional Timber Joinery Summer 2026
- Janet Qian (BS student, MIT); Topic: Topological Inverse Design of Elastic Springs Summer 2024
- Jae Yoon (David) Cha (BS student, University of Waterloo); Topic: Elastic Single Axis Joints Simulation Summer 2023
- Han Ying (BS student, CMU); Topic: Interactive Surface Parameterization Summer 2021

PROFESSIONAL SERVICE

Reviewing

SIGGRAPH, Technical Paper Tertiary Reviewer
 SIGGRAPH Asia, Technical Paper Tertiary Reviewer, Technical Communications Committee
 Symposium on Computer Animation, International Program Committee

AWARDS/HONORS

- **Merit Scholarship**, Fondation ISAE SUPAERO May '17

IT SKILLS

Programming: Python (JAX, PyTorch, TF), C++, WebGL, Matlab
Others: $\text{\LaTeX}$, Git, Google Internal Coding Infrastructure, Rhino-Grasshopper, Catia, StarCCM+, Fluent

VOLUNTEER EXPERIENCE

- **SUPAERO Junior Conseil – the Junior Entreprise of ISAE SUPAERO**, Head of the event division
- **SUPAERO Fencing Club**, President of the association
- **SUPAERO Student Association**, Section treasurer
- **Ose L'ISAE**, Volunteer in the social outreach section of ISAE SUPAERO

NON-RESEARCH WORK EXPERIENCE

Airbus Saint Eloi

Intern (laser measurements on engine pylons)

Jun '16 to Jul '16
 Toulouse, FR